Supplementary Material for "Towards Trustworthy IoT Network Security: Interpretable Open-Set Intrusion Detection"

Ngari Crisphine Macharia and Ning Weng

This supplement provides the complete concept and rule definitions, the CTU-IoT-23 feature derivation, the threshold-gradient derivation, expanded ablations, per-class and per-family results, and additional robustness analyses. The main article presents the methods, principal evidence, and limitations supporting the central conclusions. Table, figure, and section numbers here are prefixed "S"; references to the main article use its numbering.

Feature set. Following the known-class feature-selection control (Section SVIII), all CTU-IoT-23 results in the main article and in the finding-bearing supplementary tables (this control, the α-gate sweep and head-agreement, the per-family open-set breakdown, the concept-accuracy and leakage probe, the temporal split and the (λ, γ) grid) use the leak-free known-class-only feature selection; the paired comparison with the originally submitted feature list is the AUROC§ column of main Table III. Every split is drawn deterministically by the released scripts, and the released code includes a script that prints every table value from the released results.

SI. The α-Gate and Interpretability Trade-Off

We sweep λ_α over six values, five seeds each (Table S2). The regulariser raises the learned gate *monotonically*. On CTU-IoT-23, α rises from 0.38 (λ_α=0) to 0.89 (λ_α=1.0); on CIC-IoT-2023, from 0.32 to 0.95. F1 does not move: the known-class *validation* F1 (the model-selection split reported in this sweep) stays near 0.933 on CTU and 0.818–0.819 on CIC; the corresponding held-out *test* F1 is 0.933 on CTU (Table III). OOD AUROC is flat on CTU (within 0.016 across the sweep) and rises on CIC, from 0.647 at λ_α=0 to 0.658–0.673 for every $\lambda_\alpha \geq 0.05$, a gain of 0.01–0.03 (up to about one seed std) that accompanies the increase in α; as the next paragraph shows, it is not evidence that the decision moved onto the rules.

Does raising α actually move the decision onto the rules? No. A larger gate weight need not mean more rule reliance, because the two heads can emit logits of different magnitudes and the learned weights can compensate. Testing reliance directly by branch agreement (Table S1), the deployed known-class prediction (before rejection) stays with the *neural-only* head at every λ_α, and its agreement with the *rule-only* head rises only mildly as α increases (CTU 32% to 51%), staying well below neural agreement. So λ_α tunes the gate scalar but not decision-level reliance on the rules; we describe α as the learned rule-head weight in the logit blend, not as a measure of how interpretable the decision is. (This is also a global scalar gate, so it cannot route individual hard flows to the neural branch.) Each point was trained from scratch.

TABLE S1: Branch agreement across the λ_α sweep (NeSy-NIDS, mean over five seeds): learned α and the fraction of flows on which the deployed known-class prediction (before rejection) equals the neural-only (dep=neural) or rule-only (dep=rule) prediction. The open-set rejection score is computed on the hard rule vector, so these figures do not measure the rules' influence on rejection.

λ_α	0	0.05	0.1	0.2	0.5	1.0
CTU α	0.38	0.58	0.69	0.81	0.86	0.89
CTU dep=neural	1.00	1.00	1.00	1.00	1.00	1.00
CTU dep=rule	0.32	0.36	0.38	0.48	0.51	0.48
CIC α	0.32	0.81	0.86	0.90	0.93	0.95
CIC dep=neural	0.99	0.98	0.97	0.98	0.98	0.97
CIC dep=rule	0.45	0.52	0.53	0.53	0.52	0.55

TABLE S2: α-gate sweep (NeSy-NIDS, mean over five seeds): learned α and OOD AUROC per λ_α.

λ_α	0	0.05	0.1	0.2	0.5	1.0
CTU α	0.38	0.58	0.69	0.81	0.86	0.89
CTU AUROC	0.904	0.900	0.909	0.900	0.899	0.893
CIC α	0.32	0.81	0.86	0.90	0.93	0.95
CIC AUROC	0.647	0.658	0.672	0.659	0.660	0.673

SII. Concept Leakage and the γ Ablation

Table S3 reports mean concept prediction accuracy for each CBM variant over five seeds. This quantity is the *concept accuracy* of the main text: agreement between the bottleneck's predicted concept and the declared threshold label, i.e. how faithfully the model reproduces the declared predicate, not whether the predicate is semantically "correct". JointCBM without concept regularisation (γ=0) reaches 87.4% on CTU and 96.3% on CIC. On CTU, accuracy rises to 94.2% at γ=1.0 (not monotonically: 89.1% at γ=0.1, 88.9% at 0.5); at γ=0 the weakest concept is `is_persistent` at 81% (it rises to 91% by γ=1.0). On CIC it rises from 96.3% to 97.2%, with every concept above 90%. SequentialCBM reaches near-perfect concept accuracy (99.9% / 99.5%) because its concept head is trained on the concept labels alone and frozen before the classifier is trained. HybridCBM is similar (99.6% / 99.2%):

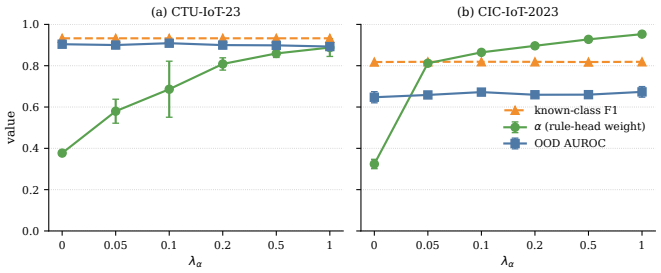


Fig. S1: α-gate sweep (NeSy-NIDS, mean over five seeds; error bars ± 1 std). Raising λ_α drives the learned gate α (the rule *weight* in the logit blend, green) monotonically upward while known-class F1 (orange, dashed) stays flat and OOD AUROC (blue) is flat on CTU-IoT-23 but rises modestly on CIC-IoT-2023.

TABLE S3: Mean concept prediction accuracy (%) on the known-class test split per CBM variant (mean over five seeds; λ sensitivity in Table S18).

Model	**CTU-IoT-23**	**CIC-IoT-2023**
JointCBM (γ=0)	87.4	96.3
JointCBM (γ=0.1)	89.1	96.5
JointCBM (γ=0.5)	88.9	96.8
JointCBM (γ=1.0)	94.2	97.2
SequentialCBM	99.9	99.5
HybridCBM	99.6	99.2

its skip connection lets the classifier bypass the bottleneck, so the concept head is under less classification pressure and can fit the concept labels more closely. Higher concept accuracy means closer agreement with the threshold labels; as the OOD results show, it does not by itself buy better detection.

Higher agreement with the labels does not by itself show that the bottleneck is free of *concept leakage*. We therefore train a linear classifier on the training split to predict the class from (i) the soft predicted concepts $\hat{\mathbf{c}} \in [0,1]^K$ and (ii) the same predictions thresholded at 0.5, and report the test-accuracy difference. This difference measures the additional predictive utility of the continuous values for these probes. It is small on CTU-IoT-23 for $\gamma \leq 0.5$ (five-seed means 0.001–0.013) but reaches $0.084 \pm .088$ at γ=1.0, where two of five seeds gain 0.17–0.21; on CIC-IoT-2023 it is 0.04–0.06 and grows rather than shrinks with γ (γ=0: $0.041 \pm .008$; γ=1.0: $0.058 \pm .008$).

A small difference does not establish semantic fidelity. Both probes use *predicted* concepts, and the thresholded predictions are not the declared predicates: where they disagree with the declared labels, the pattern of disagreement can itself carry class information. The predicate-replacement test of Section SXVI therefore uses the declared predicates directly.

SIII. Temporal (Timestamp-Ordered) Split on CTU-IoT-23

The temporal split takes the main split's per-class sample, orders each known class by capture timestamp (compared only within a class) and assigns its earliest 70% to training, the next 15% to validation and the latest 15% to test (five seeds; unknown families unchanged). Sessions or captures may straddle the boundary, so we claim temporal ordering, not capture-, session- or device-disjointness (Section SIV holds out source addresses). The CTU *DDoS* label is temporally bimodal: a high-rate flood capture (September 2018, median Rate $\approx 9.3 \times 10^5$ pps) and near-zero-rate captures (December 2018 onward, median Rate <1), so the split trains on one regime and tests on the other, and every model scores F1 0.00 on DDoS on every seed; with the main 31-feature list only two JointCBM γ=0.5 seeds keep DDoS (F1 0.65), so the cause is the label, not feature selection. Four-class F1 falls from 0.933 to 0.646–0.648 for every model but SequentialCBM (C&C-HeartBeat also falls, from 0.998 to about 0.79), a temporal-generalisation cost rather than an interpretability one. On the three coherent families the JointCBM variants, HybridCBM and NeSy-NIDS match the MLP (Coh-3 0.862–0.863) while redistributing errors (e.g. γ=0: Benign 0.841 vs. 0.958, Okiru 0.961 vs. 0.844); SequentialCBM loses Benign ($F1_4$ 0.330, Coh-3 0.441). NeSy-NIDS gives the highest temporal-split AUROC (0.888 ± 0.025; 0.901 ± 0.030 with the α penalty).

SIV. Source-IP Holdout

Flows are grouped by source IP address. Our merged flow tables keep the source IP but not the capture name, and an address can recur across captures, so a held-out address is not shown to be one physical device; the conclusions concern source addresses. CTU addresses 192.168.1.198 and 192.168.100.111 have at least 3,450 flows in every known class; each in turn supplies the known test set (3,450 per class), all other addresses supply the training and validation flows (for .198 C&C-HeartBeat has 14,699 training flows instead of 16,100), and the unknown test flows are the main split's. Features are re-selected on each split's training rows (36 for .198, 33 for .111), with the main 31-feature list (31f) as a control. The MLP, the CBM variants and NeSy-NIDS (with the α penalty and the rule-only head) are trained with five seeds (`06_make_kfs_splits.py -split device`).

TABLE S4: CTU source-IP holdout: known-class weighted F1 (five-seed mean) and OOD AUROC (mean±std, five seeds) with the re-selected features, and AUROC with the main 31-feature list (31f). With the 31-feature list, F1 is 0.417 on address .198 (0.517 for NeSy-NIDS (α)) and 0.149–0.227 on address .111. Main split for reference: F1 0.566–0.933, AUROC 0.805–0.909.

	Address 192.168.1.198			Address 192.168.100.111		
Model	F1	AUROC	AUROC 31f	F1	AUROC	AUROC 31f
MLP baseline	0.417	0.528±.020	0.529±.013	0.228	0.219±.045	0.383±.015
JointCBM γ=0	0.417	0.738±.113	0.824±.081	0.223	0.211±.102	0.111±.048
JointCBM γ=0.1	0.417	0.809±.073	0.770±.084	0.214	0.235±.085	0.113±.043
JointCBM γ=0.5	0.417	0.784±.050	0.816±.083	0.200	0.256±.088	0.114±.029
JointCBM γ=1.0	0.417	0.819±.048	0.789±.062	0.196	0.161±.015	0.093±.028
SequentialCBM	0.417	0.865±.001	0.866±.002	0.192	0.170±.013	0.202±.030
HybridCBM	0.417	0.821±.033	0.822±.041	0.152	0.215±.038	0.230±.042
NeSy-NIDS	0.417	0.843±.042	0.846±.045	0.161	0.284±.043	0.279±.044
NeSy-NIDS (α)	0.417	0.847±.045	0.847±.045	0.136	0.278±.058	0.280±.043
Rule-only	0.416	0.770±.032	0.770±.032	0.177	0.193±.035	0.193±.035

Known-class classification fails for every model. The held-out addresses' class traffic differs from the other addresses': no Benign or DDoS test flow of .198, and no test flow of .111,

has an exact same-class duplicate in training (Section SIV-A), and .198's 3,450 Benign test flows are three distinct feature vectors. With the re-selected features every MLP, CBM and NeSy-NIDS run scores F1 0.00 on .198's Benign (which the MLP labels Okiru) and on one more class, fixing weighted F1 at 0.417, and below 0.03 on .111's DDoS and Okiru (which the MLP labels C&C-HeartBeat). AUROC then no longer isolates novel attack types: on .111 it is below 0.5 in all 100 runs, because the held-out known flows look more novel than the unknown flows, over 99.9% of which come from addresses that also produce known traffic (89% for .198).

A. A Third CTU Address and TON_IoT

Address 192.168.100.108 (3,450 flows per attack class, 3,051 Benign, no unknown-family flows) gives a third CTU fold. On TON_IoT, `07_make_splits_ton.py --holdout-attacker` takes 7,500 test flows per attack class from 192.168.1.30 or .31 and 7,500 normal flows from .195 or .190, training on all other addresses; NeSy-NIDS has no TON_IoT rule bank.

TABLE S5: CTU address .108 and the two TON_IoT source-IP holdouts: known-class weighted F1 (five-seed mean) and OOD AUROC against the main split's unknown pool (mean±std, five seeds). 31f: the main 31-feature CTU list. TON_IoT main split: F1 0.704–0.898, AUROC 0.552–0.654.

	CTU address .108			TON_IoT address .30		TON_IoT address .31	
Model	F1	AUROC	AUROC 31f	F1	AUROC	F1	AUROC
MLP baseline	0.544	0.548±.003	0.542±.008	0.761	0.552±.024	0.886	0.493±.033
JointCBM γ=0	0.541	0.600±.030	0.590±.122	0.754	0.635±.007	0.885	0.585±.012
JointCBM γ=0.1	0.539	0.577±.056	0.554±.071	0.766	0.659±.027	0.853	0.597±.026
JointCBM γ=0.5	0.547	0.595±.068	0.625±.037	0.746	0.664±.023	0.858	0.613±.033
JointCBM γ=1.0	0.546	0.632±.059	0.635±.050	0.753	0.660±.017	0.888	0.617±.011
SequentialCBM	0.343	0.753±.049	0.729±.031	0.677	0.596±.009	0.731	0.601±.016
HybridCBM	0.544	0.620±.049	0.689±.077	0.767	0.610±.016	0.892	0.598±.020
NeSy-NIDS	0.552	0.907±.011	0.899±.010	–	–	–	–
NeSy-NIDS (α)	0.541	0.906±.014	0.898±.010	–	–	–	–
Rule-only	0.475	0.887±.011	0.887±.011	–	–	–	–

On TON_IoT the largest losses are on normal and password (address .30, MLP: normal 0.68 vs. 0.91, password 0.59 vs. 0.79); address .31 matches the main split. The share of known test flows with an exact same-class training duplicate (features rounded to four decimals) is 0.84 on the CTU and 0.88 on the TON_IoT main split, 0.42, 0.00 and 0.41 on CTU addresses .198, .111 and .108, and 0.74 and 0.77 on TON_IoT addresses .30 and .31.

B. Open-Set Test with Unknown Flows from the Held-Out Address

`cbm/run_device_openset.py` scores each held-out address's known test flows against that address's own unknown-family flows (MLP, CBMs, NeSy-NIDS): .198's 3,516 (3,499 C&C) and 2,000 DoS flows each from TON_IoT .30 and .31, sampled as in the main split (their 2 and 510 MITM flows are scored separately). The seen-address reference scores the main-split models against the main split's C&C (10,000 of 21,995) or DoS (2,000) flows; .111 and .108 lack unknown-family flows. TPR@5%FPR (Table S6), as in main Table III, thresholds at the 95th percentile of the held-out address's own known test scores, a diagnostic test-ROC point; Table S7 uses the deployable threshold τ (main Section III-D) set on validation flows of the training addresses.

TABLE S6: OOD AUROC / diagnostic TPR@5%FPR (five-seed mean) with known and unknown test flows from the same source address. "Seen": main-split models against the same family's flows from addresses seen in training. TON_IoT columns use DoS flows only.

	CTU, C&C			TON_IoT, DoS		
Model	Seen	Address .198	Address .198, 31f	Seen	Address .30	Address .31
MLP baseline	0.853 / 0.337	0.396 / 0.007	0.396 / 0.007	0.444 / 0.009	0.411 / 0.004	0.818 / 0.016
JointCBM γ=0	0.829 / 0.339	0.609 / 0.052	0.713 / 0.162	0.754 / 0.016	0.691 / 0.007	0.749 / 0.016
JointCBM γ=0.1	0.832 / 0.331	0.663 / 0.071	0.688 / 0.066	0.735 / 0.011	0.705 / 0.003	0.757 / 0.024
JointCBM γ=0.5	0.843 / 0.439	0.637 / 0.020	0.712 / 0.100	0.725 / 0.014	0.750 / 0.002	0.792 / 0.026
JointCBM γ=1.0	0.854 / 0.507	0.690 / 0.073	0.665 / 0.030	0.735 / 0.011	0.703 / 0.003	0.772 / 0.001
SequentialCBM	0.878 / 0.497	0.635 / 0.039	0.638 / 0.041	0.719 / 0.022	0.693 / 0.000	0.691 / 0.002
HybridCBM	0.876 / 0.554	0.636 / 0.033	0.637 / 0.032	0.722 / 0.021	0.690 / 0.001	0.645 / 0.011
NeSy-NIDS	0.865 / 0.260	0.651 / 0.060	0.651 / 0.060	–	–	–
NeSy-NIDS (α)	0.872 / 0.247	0.651 / 0.066	0.650 / 0.057	–	–	–
Rule-only	0.820 / 0.183	0.639 / 0.031	0.639 / 0.031	–	–	–

TABLE S7: Deployable threshold τ (95th percentile of known validation scores): share of known test flows rejected / recall of unknown flows (five-seed mean). The target rejection rate is 5%.

	CTU, C&C			TON_IoT, DoS		
Model	Seen	Address .198	Address .198, 31f	Seen	Address .30	Address .31
MLP baseline	0.049 / 0.337	0.501 / 0.049	0.501 / 0.049	0.050 / 0.009	0.070 / 0.004	0.131 / 0.070
JointCBM γ=0	0.048 / 0.276	0.250 / 0.028	0.051 / 0.038	0.052 / 0.016	0.062 / 0.008	0.060 / 0.016
JointCBM γ=0.1	0.051 / 0.335	0.151 / 0.027	0.201 / 0.033	0.051 / 0.016	0.057 / 0.003	0.081 / 0.037
JointCBM γ=0.5	0.052 / 0.439	0.200 / 0.035	0.151 / 0.037	0.051 / 0.014	0.060 / 0.005	0.058 / 0.047
JointCBM γ=1.0	0.053 / 0.513	0.126 / 0.131	0.200 / 0.033	0.051 / 0.017	0.070 / 0.006	0.077 / 0.003
SequentialCBM	0.055 / 0.549	0.000 / 0.031	0.000 / 0.032	0.051 / 0.021	0.099 / 0.003	0.062 / 0.002
HybridCBM	0.050 / 0.552	0.050 / 0.026	0.100 / 0.027	0.051 / 0.021	0.088 / 0.004	0.037 / 0.002
NeSy-NIDS	0.047 / 0.286	0.030 / 0.042	0.022 / 0.041	–	–	–
NeSy-NIDS (α)	0.047 / 0.268	0.016 / 0.040	0.023 / 0.040	–	–	–
Rule-only	0.039 / 0.183	0.177 / 0.045	0.177 / 0.045	–	–	–

On CTU the held-out address lowers C&C AUROC from 0.82–0.88 to 0.40–0.71 and diagnostic TPR from 0.18–0.55 to at most 0.162; on TON_IoT address .30 the change is small (AUROC 0.41–0.75 vs. 0.44–0.75), and TPR is low on seen addresses too ($\leq$ 0.022). The deployable threshold keeps known rejection near target on seen addresses (3.9–5.5%) but not on held-out ones (0–50% on .198, 3.7–13.1% on TON_IoT), with unknown recall at most 13.1% (.198) and 7.0% (TON_IoT). The model ordering is not stable across addresses (the concept models are above the MLP on .198 and .30, below it on .31). On the 510 MITM flows of .31, AUROC is 0.63–0.85 and diagnostic TPR at most 0.139.

SV. Confidence Intervals

`cbm/run_bootstrap_ci.py` recomputes the per-flow OOD scores of every model compared from its saved checkpoints (checked against the stored AUROC) and draws 2,000 hierarchical bootstrap replicates: the five seeds are resampled with replacement for each model independently, and the 5,000 known and 5,000 unknown evaluation flows are resampled with replacement and shared by the two models of a comparison, since every model is scored on the same flows. The interval is the 2.5–97.5 percentile of the difference of the two seed-mean AUROCs. The intervals cover the variation from training seeds and from the sampled evaluation flows under the chosen splits; they do not measure uncertainty across new captures, source addresses or independently drawn splits.

Every difference the main text describes as a gap excludes zero. An interval that includes zero shows that this analysis

TABLE S8: Differences in mean OOD AUROC with 95% bootstrap confidence intervals.

Data	Comparison	Difference	95% CI
CTU	NeSy-NIDS − MLP	+0.030	[+0.016, +0.043]
CTU	NeSy-NIDS (α) − MLP	+0.035	[+0.020, +0.047]
CTU	SequentialCBM − MLP	−0.002	[−0.027, +0.018]
CTU	JointCBM γ=1.0 − JointCBM γ=0	+0.011	[−0.052, +0.065]
CTU	JointCBM γ=0 − 8-unit bottleneck	+0.263	[+0.102, +0.380]
CIC	MLP − JointCBM γ=0	+0.090	[+0.073, +0.108]
CIC	JointCBM γ=1.0 − JointCBM γ=0	−0.018	[−0.037, +0.002]
CIC	8-unit bottleneck − JointCBM γ=0	+0.003	[−0.024, +0.027]
CIC	NeSy-NIDS, energy score − NeSy-NIDS, Mahalanobis	+0.073	[+0.035, +0.112]
TON	JointCBM γ=0 − MLP	+0.098	[+0.051, +0.145]
TON	JointCBM γ=0 − 8-unit bottleneck	+0.106	[+0.004, +0.214]

does not resolve the difference, not that the two models are equivalent; the main text describes those comparisons as similar observed AUROC or as no consistent improvement. The TON_IoT difference between JointCBM and the 8-unit bottleneck barely excludes zero, because the bottleneck varies widely across seeds.

SVI. Rule Selectivity and Threshold Drift

All values in this section are mean binarised activations (Eq. 5) per known class on the known-class validation split for the seed-0 model, the same data as Fig. 2. On CTU-IoT-23, several rules fire selectively but not deterministically (`cc_heartbeat_beaconing` at 0.78 and `incomplete_handshake` at 0.87 on C&C-HeartBeat, `asymmetric_flood` at 0.81 on DDoS and 0.13 on Benign, `ddos_high_rate` at 0.70 on DDoS and 0.21 on Okiru), leaving within-class variance for Mahalanobis to work with; others fire across several classes (`okiru_low_rate`: 0.79 on Okiru but 0.87 on Benign and 0.61 on C&C-HeartBeat). On CIC-IoT-2023 firing is near-deterministic (`syn_flood_ddos` at 1.00 on DDoS-SYN, `benign_low_rate` at 0.97 on Benign but also 0.91 on Mirai), within-class variance is close to zero, and Mahalanobis has little to separate on (AUROC 0.647). These two activation regimes coincide with the observed CTU-versus-CIC difference in Mahalanobis AUROC, but, as the main paper establishes, known-class activation geometry does not by itself govern detectability, so we report the pattern as descriptive, not causal (Fig. 2). The learned thresholds move away from their expert initial values (seed 0): on CTU-IoT-23 the `cc_heartbeat_beaconing` bounds move by up to −0.50, and on CIC-IoT-2023 by up to −0.88 on `syn_flood_ddos`; across the five seeds the per-condition drift vectors are correlated on CIC-IoT-2023 (mean pairwise r=0.73) but not on CTU-IoT-23 (r=0.05; script `nesy/run_fidelity.py`), so we read the CIC refinement as systematic and the CTU pattern as indicative only (main Fig. 3).

SVII. Configuration Sensitivity of the Geometry Correlations

Main Table XIII correlates each known-class statistic with OOD AUROC over 30 CBM models per dataset, which are five seeds of six configurations rather than 30 independent observations. Table S9 reports, on the same leak-free feature lists as the main table (CTU: known-class-only selection, Section SVIII; CIC/TON: fixed lists), two further aggregations applied identically to all three corpora: the correlation over the six configuration means (r_6, n=6), and the range of the 30-model correlation when each configuration in turn is excluded (leave-one-configuration-out, LOCO). On CIC-IoT-2023 and TON_IoT the seven statistics agree in direction, and the six separability and variance statistics are strongly associated with AUROC at the configuration-mean level ($|r_6| \geq 0.93$ on both); closed-set F1 is weaker (r_6=0.63 and 0.73) and is the only statistic whose sign changes under LOCO there (on CIC). CTU-IoT-23 is different. All seven statistics take the opposite sign (r_6 from −0.82 to +0.49), and the LOCO ranges show why: every CTU range reaches its bound closest to zero when SequentialCBM is excluded, where the correlations fall to $|r| \leq 0.22$ and Fisher ratio, silhouette and closed-set F1 change sign. SequentialCBM combines the least separable concept space and the lowest closed-set F1 with one of the highest CTU AUROCs, so this single configuration sets the CTU direction. The unknown/known ratio, which uses the unknown flows, is positive everywhere (r_6 0.70–0.92). These are associations with Mahalanobis AUROC measured *within* each corpus, and the strong r_6 values rest on six points. Prediction on a held-out corpus and scorer selection are tested separately (Sections SXII and SXIII), and neither succeeds. The LOCO analysis thus locates the CTU reversal in one configuration rather than in the corpus.

TABLE S9: Configuration sensitivity of main Table XIII, on the same leak-free feature lists (CTU: known-class-only selection). r_6: Pearson r over the six configuration means. LOCO: range of the 30-model r when one configuration is excluded; on CTU the bound nearest zero is always the run without SequentialCBM. *Sign changes when SequentialCBM is excluded.

	CTU-IoT-23		CIC-IoT-2023		TON_IoT	
Statistic	r_6	LOCO	r_6	LOCO	r_6	LOCO
Within-class variance	+0.49	[+0.01, +0.42]	−0.96	[−0.82, −0.66]	−0.95	[−0.54, −0.32]
Fisher ratio	−0.45	[−0.32, +0.02]*	+0.96	[+0.56, +0.76]	+0.95	[+0.36, +0.54]
Ridge-LDA	−0.50	[−0.44, −0.08]	+0.95	[+0.31, +0.60]	+0.98	[+0.40, +0.57]
Silhouette	−0.46	[−0.42, +0.01]*	+0.95	[+0.74, +0.85]	+0.94	[+0.40, +0.58]
Nearest-centroid margin	−0.47	[−0.43, −0.02]	+0.94	[+0.50, +0.74]	+0.94	[+0.32, +0.52]
Concept entropy	−0.52	[−0.47, −0.06]	+0.96	[+0.71, +0.83]	+0.93	[+0.25, +0.49]
Closed-set F1	−0.82	[−0.42, +0.22]*	+0.63	[−0.28, +0.76]*	+0.73	[+0.03, +0.42]
Unknown/known ratio	+0.78	[+0.59, +0.74]	+0.92	[+0.43, +0.64]	+0.70	[+0.70, +0.76]

SVIII. CTU-IoT-23 Feature Derivation

CTU-IoT-23 is distributed as packet captures with labelled Zeek `conn.log` files. For each capture we compute per-flow packet statistics with DPKT (field names follow the CIC-IoT-2023 feature set) and join each labelled Zeek record to the DPKT flows with the same source and destination IP, ports and protocol whose start time lies in $[t_{\text{zeek}} - \text{duration} - 60\,\text{s},\ t_{\text{zeek}} + 60\,\text{s}]$, keeping the one closest in start time (`preprocess/merge_iot23_outer2.py`), so each Zeek record yields at most one row; in the stored table only two pairs of rows share source address, timestamp, duration, byte counts, history and label. Zeek records without a DPKT match are kept, with their DPKT features set to 0: 1,896,696 of the 110,879,826 retained flows (1.7%), mostly DDoS (7.6% of DDoS flows); no flow matched with protocol ICMP. In the main split they are 2.2% of the training and

2.0% of the known test flows (mostly DDoS) and 0.4% of the unknown flows. Before sampling, a label filter excludes PartOfAHorizontalPortScan, PartOfAHorizontalPortScan-Attack and C&C-PartOfAHorizontalPortScan, so horizontal port scanning is neither a known nor a held-out class. We used the full IoT-23 archive (`iot_23_datasets_full.tar.gz`), whose packet captures the DPKT features require, and keep the labels of its Zeek label files. For capture CTU-IoT-Malware-Capture-33-1 these files label 13,609,467 flows Okiru, a count the provider's summary table lists under Okiru-Attack; our flows from that capture's source address (192.168.1.197) include exactly 13,609,467 Okiru and no Okiru-Attack flows, and the label file for this capture currently served by the provider also uses Okiru. These flows therefore belong to the known Okiru class, and our three Okiru-Attack flows are those of capture 36-1; Table S10 lists the retained labels (`checks/ctu_label_counts.py`).

TABLE S10: Retained CTU-IoT-23 flows per label and those without a matching DPKT flow (DPKT features set to 0).

Label	Retained	No DPKT match
Okiru	60,717,929	267,008
Benign	30,830,238	158,334
DDoS	19,265,778	1,471,222
C&C-HeartBeat	33,537	6
C&C	21,995	103
Attack	9,398	1
C&C-HeartBeat-Attack	834	18
C&C-FileDownload	53	4
C&C-Torii	30	0
FileDownload	18	0
C&C-HeartBeat-FileDownload	11	0
Okiru-Attack	3	0
C&C-Mirai	2	0

Four ratios are derived from Zeek fields: byte ratio $\mathrm{orig_bytes}/(\mathrm{resp_bytes}+1)$, originator packet and byte rates $\mathrm{orig_pkts}/(\mathrm{duration}+1)$ and $\mathrm{orig_ip_bytes}/(\mathrm{duration}+1)$, and direction $\mathrm{orig_bytes}-\mathrm{resp_bytes}$. The model uses 34 numeric features (Zeek: duration, orig/resp bytes, packets and IP bytes; DPKT: Rate, Srate, IAT, Number, Tot_sum, Min, Max, AVG, Std, Variance, Magnitude, Radius, ack/syn/fin/rst counts, fin/rst flag numbers, and HTTP, HTTPS, SSH, IRC, Telnet indicators; plus the four derived ratios) and three Zeek categorical fields (conn_state, history, service) integer-encoded, 37 in total. Missing values are set to -1; features are not rescaled.

Feature-selection control. The 34 numeric features were chosen before this study by a class-weighted random forest (99% of cumulative importance) trained on up to 10^6 flows of every labelled CTU family, including the nine later held out, so the submitted list is not blind to the held-out labels (CIC-IoT-2023 and TON_IoT use fixed lists). We therefore re-ran the same selection on the known-class training rows only: the resulting list keeps 28 of the numeric features (it drops HTTP, HTTPS, SSH, IRC, `resp_pkts`, `fin_count`, Radius, Std, Variance, `resp_ip_bytes` and `rst_flag_number`, and adds TCP, Protocol_Type, Time_To_Live and the `cwr` and `syn` flag numbers), 31 features with the three categorical fields (`ctuk`). The submitted list on the same rows (`ctuko`) isolates the selection source from the sample. NeSy-NIDS keeps its expert rule features, so the two rule counters the known-only forest drops are restored (`06_make_kfs_splits.py`, `cbm/run_kfs_control.py`, `nesy/run_kfs_control.sh`). Main Table III reports the outcome over five seeds (F1; AUROC[§] for `ctuko`). Known-class F1 is unchanged; per-class Mahalanobis remains the best CTU scorer for every architecture (Table S11); γ still has no monotone effect. All OOD AUROC changes of 0.05 or more are gains under known-class-only selection (SequentialCBM $0.782\pm.042\rightarrow0.872\pm.026$, HybridCBM $0.772\pm.073\rightarrow0.822\pm.055$, Post-hoc CBM $0.805\rightarrow0.871$); the MLP moves -0.020, NeSy-NIDS $+0.019$ ($0.885\rightarrow0.904$). The submitted list therefore did not flatter the CTU detection results.

TABLE S11: Scorer comparison under the known-only feature selection (`ctuk`): OOD AUROC (mean±std, five seeds) for per-class Mahalanobis on the concept space, the logit-space energy score, and their average after z-normalising each with known validation scores (combined). Per-class Mahalanobis is best for every architecture, and the logit-energy score reaches at most 0.61 (main Table VII gives the CIC comparison, where the ranking depends on architecture).

Architecture	Mahalanobis	Energy	Combined
MLP	**0.879±.001**	0.386±.073	0.682±.046
JointCBM (γ=0)	**0.830±.044**	0.476±.111	0.631±.068
JointCBM (γ=0.1)	**0.812±.034**	0.494±.136	0.624±.098
JointCBM (γ=0.5)	**0.851±.037**	0.570±.138	0.714±.106
JointCBM (γ=1)	**0.842±.049**	0.612±.124	0.708±.100
SequentialCBM	**0.875±.026**	0.604±.085	0.757±.018
HybridCBM	**0.827±.055**	0.365±.047	0.610±.047

Per-split selection. The same selection is re-run separately for each other CTU experiment: on the alternative known set's training rows, the earliest 70% of the temporal split and each leave-one-family-out fold's reduced known set, so no held-out family or test-period flow informs its features. The alternative split draws its own class-balanced sample, the temporal split reorders the main split's sample by time, and the leave-one-family-out folds reuse the main split's rows; for the alternative and temporal splits, the main 31-feature list on the same rows is a control, and the accuracy, scorer and γ conclusions hold under both.

SIX. NeSy-NIDS Rule Bank and Initial Thresholds

Table S12 gives the complete eight-rule bank for each dataset used by NeSy-NIDS (Section III-C of the main paper lists only representative rules). Each rule is a conjunction of threshold conditions on named flow features; the thresholds shown are the expert *initial* values $\boldsymbol{\theta}^{(0)}$, which are then refined by gradient descent through the k-annealed gates. The full feature lists (per-dataset `feature_cols`), the deterministic concept/rule threshold values, and the train/val/test split indices are released with the code (see the code-availability statement), so every representation in the paper is reproducible.

TABLE S12: Complete NeSy-NIDS rule bank (R=8 per dataset): target class and the conjunction of conditions at the expert *initial* thresholds (refined during training). CTU-IoT-23 in raw units; CIC-IoT-2023 in z-scored units.

Rule	Target	Condition (initial threshold)
CTU-IoT-23 (raw units)		
ddos_high_rate	DDoS	Rate>1000
cc_heartbeat_beaconing	C&C-HB	1.5<IAT<5.0 ∧ 0.1<Rate<5.0
benign_single_probe	Benign	orig_pkts<2∧ Rate<100
okiru_low_rate	Okiru	Rate<500∧ duration<2.5
asymmetric_flood	DDoS	resp_pkts<1∧ orig_pkts>1
persistent_connection	C&C-HB	duration>2.5
incomplete_handshake	C&C-HB	syn_count>2∧ fin_count<1
ddos_short_burst	DDoS	Rate>1000∧ duration<0.1
CIC-IoT-2023 (z-scored units)		
syn_flood_ddos	DDoS-SYN	rate>0.15∧ syn_count>1.5
udp_mirai_flood	Mirai	tcp<−1.0∧ rate>0.0
recon_port_scan	Recon	rst_count>1.5
vuln_scan	VulnScan	fin_count>1.0∧ rate>0.0
benign_low_rate	Benign	rate<0.0∧ syn_count<0.0
large_payload_flood	Benign	avg>0.4∧ udp>0.0
high_entropy_scan	Recon	variance>0.5∧ rst_count>0.5
short_connection_flood	DDoS-SYN	iat<−0.003∧ rate>0.1

SX. Threshold Gradient

Training uses the soft activations of main Eq. 4 throughout, so every threshold receives a gradient. For a greater-than condition $i \in \mathcal{I}_r^{>}$ of rule r,

$$\frac{\partial \mathcal{L}}{\partial \theta_i} = \frac{\partial \mathcal{L}}{\partial a_r} \cdot \Big(-k(t)\, g_i^{>} (1 - g_i^{>}) \prod_{j \neq i} g_j \Big), \tag{1}$$

where $g_i^{>} = \sigma(k(t)(x_i - \theta_i))$ and g_j are the other gates of the rule; the sign flips for less-than conditions. As $k(t)$ grows, $g_i^{>}(1-g_i^{>})$ vanishes for gates far from their threshold, so late in training only borderline conditions keep moving. The straight-through estimator is not used in training.

SXI. Scorer Settings

Table S13 lists how each OOD score is computed. Every setting was fixed in advance (the ODIN and kNN values are the published defaults) and none was chosen using the held-out families; scorers are fitted on training vectors, and the combined score is normalised with known validation scores, so no test flow enters any scorer.

SXII. Leave-Family-Out Scorer Selection

The measured associations between known-class statistics and OOD AUROC vary across datasets (main Table XIII); whether a scorer can be *selected* in advance is a separate question, which we test directly with a *behavioural* proxy: hold out one known attack family, retrain JointCBM (γ=0) on the reduced known set, and keep the scorer (per-class Mahalanobis vs logit-space energy) that best detects the held-out family, then ask whether that choice matches the scorer actually best on the true unknowns (Mahalanobis on CTU-IoT-23, energy on CIC-IoT-2023). We repeat for every known attack family × five seeds (main Table VIII).

TABLE S13: OOD scorer settings.

Scorer	Computation
Mahalanobis	Per-class means and a shared covariance of the within-class residuals ($+10^{-4}I$, pseudo-inverse), fitted on 20,000 training vectors; score = distance to the nearest class mean
Hamming, Jaccard	Hard rule vectors; each class representative is the majority vote of its training vectors (rule mean > 0.5, so it is binary); score = Hamming or Jaccard distance to the nearest representative
Energy	$-\log \sum_c e^{l_c}$ on the logits, T=1
Combined	Mahalanobis and energy scores each z-normalised with the mean and standard deviation of the known validation scores (frozen before testing), then summed
MSP	$1 - \max_c \mathrm{softmax}(l)_c$
ODIN	MSP at temperature T=1000 after an input perturbation of ϵ=0.0014 along the sign of the gradient
kNN	Euclidean distance to the 50th nearest L_2-normalised training embedding (20,000 training vectors)

The proxy is only *conditionally* reliable. On CTU-IoT-23 the family-majority vote recovers Mahalanobis (two of three families, 9/15 runs); the C&C-HeartBeat and DDoS held-out-family experiments favour it, and only Okiru misfires (Mahalanobis cannot detect held-out Okiru, 0.22). On CIC-IoT-2023 the family-majority vote selects Mahalanobis for all four families (15/20 runs, 5/20 energy) and so never recovers energy: held-out known families remain separable in activation space (Mahalanobis 0.70–0.86), so the proxy selects Mahalanobis, yet the true CIC unknowns overlap the known boundary and need energy. Held-out-known thus over-approximates activation-space detectability exactly on the corpus where the scorer must switch. In these experiments, therefore, this leave-family-out proxy did not consistently identify the best scorer; selecting it reliably would require observing the novel regime, which we do not demonstrate a priori. This is a further instance of the paper's negative result, here for scorer *selection* rather than scorer performance. Driver: `cbm/run_lofo.py`. A second rule that needs no retraining, based on synthetic outliers, is evaluated in Section SXIII.

SXIII. Transfer and Selection Tests

Transfer across corpora. For each known-class statistic s we fit AUROC $= a + b\,s$ by least squares on the 60 CBM concept-space models of two corpora and apply it to the 30 models of the third (Table S14; `cbm/run_geometry_transfer.py`). The prediction beats the training-corpora mean AUROC in 3 of 21 cases, ranks the held-out models in the right direction in 7 of 21, and identifies the best configuration in 1 of 21. The slope is dominated by between-corpus differences (the dataset means are non-monotone), so it often has the wrong sign even for CIC-IoT-2023 and TON_IoT, whose within-corpus directions agree.

Scorer selection from synthetic outliers. As a second known-traffic rule that needs no retraining, we select Mahalanobis or energy by which better separates validation flows from synthetic outliers: feature-wise permutations of validation flows (primary, fixed in advance) or midpoints of random pairs (Table S15; `cbm/run_scorer_selection.py`). Against

TABLE S14: Leave-one-corpus-out prediction of OOD AUROC from each known-class statistic (fitted on the other two corpora, 60 models; evaluated on the 30 held-out models). Slope: fitted b; MAE: mean absolute AUROC error on the held-out corpus; ρ: Spearman correlation of predicted and true AUROC; Best: the predicted best configuration is the true best.

Statistic	Slope	MAE	ρ	Best
Held out: CTU (training-mean MAE 0.167)				
Within-class variance	−1.774	0.198	−0.36	no
Fisher ratio	+0.065	0.169	−0.34	no
Ridge-LDA	+0.000	0.180	−0.38	no
Silhouette	+0.336	0.149	−0.34	no
Nearest-centroid margin	+0.201	0.152	−0.28	no
Concept entropy	+0.006	0.167	−0.38	no
Closed-set F1	−0.022	0.168	+0.17	yes
Held out: CIC (training-mean MAE 0.045)				
Within-class variance	+0.616	0.030	−0.66	no
Fisher ratio	+0.060	0.075	+0.60	no
Ridge-LDA	+0.003	0.671	+0.41	no
Silhouette	+0.263	0.069	+0.75	no
Nearest-centroid margin	+0.247	0.083	+0.54	no
Concept entropy	−0.415	0.064	−0.70	no
Closed-set F1	−0.006	0.045	−0.17	no
Held out: TON (training-mean MAE 0.123)				
Within-class variance	+2.001	0.146	−0.46	no
Fisher ratio	−0.045	0.145	−0.51	no
Ridge-LDA	−0.000	0.153	−0.58	no
Silhouette	−0.092	0.136	−0.43	no
Nearest-centroid margin	−0.113	0.149	−0.39	no
Concept entropy	+0.113	0.130	+0.37	no
Closed-set F1	+0.128	0.125	+0.30	no

the scorer better on the true unknowns, the permutation rule is right for $25/35$ CTU, $26/35$ CIC and $9/35$ TON_IoT models ($60/105$), the midpoint rule for $27/35$, $21/35$ and $9/35$ ($57/105$); always choosing Mahalanobis is right for $62/105$ and energy for $43/105$, and wrong permutation choices cost 0.066 AUROC on average. Both rules prefer Mahalanobis on TON_IoT, where energy is better for every architecture.

SXIV. Open-Set (C+1) and Per-Family Evaluation

Protocol. The deployed $(C{+}1)$-class detector rejects a flow as unknown if its per-class Mahalanobis score (concept activations for the CBMs, hard rule activations for NeSy-NIDS, the embedding for the MLP) exceeds τ, the 95th percentile on known validation flows, and otherwise takes the $\arg\max$ known-class logit. It is evaluated on $5{,}000$ known test flows and up to $3{,}000$ flows per held-out family, all families merged into one unknown class (9 on CTU, 29 on CIC; main Table IX).

The known rejection rate is close to target (4.2–5.8%; the coefficient of variation of τ across seeds is at most 0.11). Open-set macro-F1 is 0.49–0.79 on CTU-IoT-23 and 0.13–0.27 on CIC-IoT-2023, largely a prevalence effect. For NeSy-NIDS on CIC-IoT-2023 (Table S16, seed 0) the pool has $5{,}000$ known and $84{,}514$ unknown flows, of which $15{,}646$ (18.5%) exceed τ; the $\sim$81% not rejected are absorbed into known classes, and since unknowns outnumber knowns $\sim$17:1 they collapse known-class precision (Benign ≈ 0.07, F1 ≈ 0.13), while the unknown class has the highest per-class F1 (≈ 0.31). Reweighting the same matrix to equal known and unknown totals, with all conditional rates unchanged, raises macro-F1 from ≈ 0.14 to ≈ 0.51, an arithmetic check rather than a better model. We therefore base the detection claim on the prevalence-free per-family AUROC (Table S17).

TABLE S15: Scorer selection from synthetic outliers (five seeds). True: OOD AUROC on the held-out unknown families; Perm.: AUROC separating validation flows from feature-permuted copies; Correct: seeds on which the rule picks the scorer that is better on the true unknowns (permutation / midpoint rule).

	True		Perm.		Correct	
Architecture	Mahal.	Energy	Mahal.	Energy	Perm.	Mid.
CTU-IoT-23						
MLP	0.879	0.386	0.984	0.766	5/5	4/5
JointCBM (γ=0)	0.830	0.476	0.802	0.703	4/5	4/5
JointCBM (γ=0.1)	0.812	0.494	0.795	0.747	3/5	4/5
JointCBM (γ=0.5)	0.851	0.570	0.823	0.747	4/5	5/5
JointCBM (γ=1.0)	0.842	0.612	0.853	0.786	5/5	5/5
SequentialCBM	0.875	0.604	0.644	0.585	4/5	4/5
HybridCBM	0.827	0.365	0.651	0.801	0/5	1/5
CIC-IoT-2023						
MLP	0.807	0.615	0.974	0.655	5/5	5/5
JointCBM (γ=0)	0.722	0.734	0.769	0.803	3/5	3/5
JointCBM (γ=0.1)	0.705	0.732	0.779	0.821	4/5	2/5
JointCBM (γ=0.5)	0.706	0.757	0.771	0.826	5/5	1/5
JointCBM (γ=1.0)	0.702	0.769	0.778	0.820	5/5	0/5
SequentialCBM	0.665	0.636	0.723	0.689	4/5	5/5
HybridCBM	0.670	0.629	0.678	0.732	0/5	5/5
TON_IoT						
MLP	0.550	0.662	0.859	0.523	0/5	0/5
JointCBM (γ=0)	0.644	0.713	0.740	0.621	1/5	0/5
JointCBM (γ=0.1)	0.640	0.674	0.758	0.624	3/5	2/5
JointCBM (γ=0.5)	0.641	0.694	0.770	0.630	2/5	2/5
JointCBM (γ=1.0)	0.642	0.706	0.758	0.637	1/5	1/5
SequentialCBM	0.603	0.640	0.730	0.637	0/5	2/5
HybridCBM	0.615	0.641	0.747	0.548	2/5	2/5

TABLE S16: $(C{+}1)$ confusion matrix, NeSy-NIDS on CIC-IoT-2023 (seed 0; row = true, column = predicted). All 29 held-out families are merged into the single "unknown/reject" row. VulnSc.: VulnerabilityScan.

true \ pred	Benign	DDoS-SYN	Mirai	Recon	VulnSc.	reject
Benign	765	0	0	70	161	1
DDoS-SYN	0	911	0	0	0	37
Mirai	0	0	1073	0	0	3
Recon-PortScan	155	0	0	349	265	175
VulnScan	126	0	0	95	801	13
unknown	9633	13459	16918	6349	22509	15646

Averaging the recall column of Table S17 over families gives the family-averaged recall of main Table IX (0.51/0.36 for JointCBM, 0.83/0.20 for NeSy-NIDS on CTU/CIC). The equal-family mean AUROC (0.818 CTU, 0.722 CIC for JointCBM) is close to the pooled AUROC of main Table III (0.823, 0.717), but its per-family detection ranges from near-perfect (several volumetric DDoS families, AUROC > 0.97) to below chance (CIC Mirai-greip_flood 0.15, DDoS-UDP_Fragmentation 0.24), is weak for most injection, scan and web families (0.58–0.71, recall $\leq$0.22), and differs between the two models.

TABLE S17: Per-family OOD AUROC and unknown recall at τ (5% FPR) for JointCBM (γ=0) and NeSy-NIDS, mean over five seeds. n: evaluated flows of the family, the same for every seed (sampled without replacement, at most 3,000; each a distinct dataset record), so the five seeds do not add test flows and rare CTU families rest on very few flows. Detection varies enormously by family for both models, and the two disagree on which families they catch.

		JointCBM		NeSy-NIDS	
Held-out family	n	AUROC	Rec.	AUROC	Rec.
CTU-IoT-23					
C&C	3,000	0.832	0.33	0.864	0.29
Attack	3,000	0.812	0.45	0.995	0.99
C&C-HeartBeat-Attack	834	0.819	0.55	0.993	0.94
C&C-FileDownload	53	0.712	0.42	0.996	1.00
C&C-Torii	30	0.812	0.27	0.844	0.28
C&C-Mirai	2	0.994	1.00	0.994	1.00
C&C-HeartBeat-FileDownload	11	0.701	0.40	0.995	1.00
FileDownload	18	0.715	0.40	0.995	1.00
Okiru-Attack	3	0.964	0.80	0.979	0.93
CIC-IoT-2023					
DDoS-UDP_Flood	3,000	0.978	0.847	0.888	0.453
DDoS-TCP_Flood	3,000	0.902	0.390	0.830	0.107
DDoS-ICMP_Flood	3,000	0.949	0.638	0.900	0.481
DDoS-RSTFINFlood	3,000	0.974	0.816	0.993	0.958
DDoS-PSHACK_Flood	3,000	0.993	0.993	0.848	0.178
DDoS-SynonymousIP_Flood	3,000	0.475	0.002	0.350	0.015
DDoS-ACK_Fragmentation	3,000	0.962	0.821	0.516	0.055
DDoS-UDP_Fragmentation	3,000	0.237	0.024	0.892	0.343
DDoS-HTTP_Flood	3,000	0.969	0.881	0.868	0.606
DDoS-SlowLoris	3,000	0.977	0.898	0.705	0.284
DDoS-ICMP_Fragmentation	3,000	0.636	0.219	0.459	0.007
DoS-SYN_Flood	3,000	0.543	0.088	0.452	0.124
DoS-UDP_Flood	3,000	0.970	0.761	0.839	0.365
DoS-TCP_Flood	3,000	0.883	0.333	0.809	0.077
DoS-HTTP_Flood	3,000	0.975	0.907	0.897	0.697
Mirai-greip_flood	3,000	0.148	0.021	0.551	0.005
Mirai-udpplain	3,000	0.632	0.029	0.937	0.628
Recon-HostDiscovery	3,000	0.711	0.223	0.560	0.098
Recon-OSScan	3,000	0.661	0.066	0.537	0.074
Recon-PingSweep	2,262	0.635	0.091	0.454	0.014
MITM-ArpSpoofing	3,000	0.742	0.245	0.585	0.072
DNS_Spoofing	3,000	0.721	0.291	0.547	0.060
DictionaryBruteForce	3,000	0.582	0.065	0.441	0.007
BrowserHijacking	3,000	0.663	0.117	0.496	0.026
CommandInjection	3,000	0.647	0.156	0.487	0.008
XSS	3,000	0.597	0.103	0.441	0.006
SqlInjection	3,000	0.577	0.077	0.421	0.003
Backdoor_Malware	3,000	0.607	0.120	0.446	0.006
Uploading_Attack	1,252	0.596	0.128	0.431	0.004

SXV. Joint (λ, γ) Sweep

λ (BCE concept supervision) and γ (added L2 concept fidelity) both push the predicted concepts toward the same labels, so we sweep them jointly (JointCBM, five seeds per cell, main splits, evaluation flows of main Table III; the λ=0.5 row is the JointCBM rows of that table) rather than as two 1-D slices (Table S18). Three points hold in 2-D. (i) γ is not a reliable lever: its effect $\gamma_{1.0}-\gamma_0$ is at most 0.040 and changes with λ, growing from -0.004 at λ=0.1 to $+0.030$ at λ=1.0 on CTU-IoT-23 and shrinking from -0.040 to -0.003 on CIC-IoT-2023. (ii) λ moves AUROC by a similar amount in opposite directions on the two corpora (down the γ=0 column, $+0.020$ on CTU, -0.040 on CIC). (iii) Known-class F1 is nearly constant across the grid (0.816–0.821 on CIC; 0.932–0.933 on CTU) except for two CTU runs at λ=1.0 (γ=0 seed 1 and γ=1.0 seed 2), where F1 falls to 0.734 and 0.736; both are *included* in the cell means, and they show that strong concept supervision can destabilise training. No single tunable factor governs open-set detectability.

TABLE S18: Joint (λ, γ) sweep: JointCBM OOD AUROC on the main splits (mean over five seeds; std in parentheses), λ down rows, γ across columns; F1 and concept accuracy (Conc.) at γ=0. The γ=0 column is the λ sensitivity, and the λ=0.5 row is the JointCBM rows of main Table III.

	γ=0	γ=0.5	γ=1.0	F1	Conc.
CTU-IoT-23					
λ=0.1	0.810 (.021)	0.801 (.040)	0.806 (.047)	0.9331	0.700
λ=0.5	0.823 (.044)	0.844 (.036)	0.834 (.050)	0.9330	0.874
λ=1.0	0.830 (.041)	0.854 (.029)	0.860 (.037)	0.8931	0.935
CIC-IoT-2023					
λ=0.1	0.743 (.018)	0.724 (.023)	0.703 (.020)	0.8196	0.865
λ=0.5	0.717 (.017)	0.702 (.007)	0.699 (.013)	0.8183	0.963
λ=1.0	0.703 (.013)	0.704 (.005)	0.700 (.007)	0.8184	0.975

SXVI. Concept-Fidelity Intervention (CBM)

Because our concepts are deterministic functions of the input, we test how much JointCBM's accuracy depends on deviations from the declared predicates: we (i) run the model normally ($F1_{pred}$), (ii) replace the predicted concepts with the true binary predicates and pass them through the model's own classifier head ($F1_{gt}$), and (iii) fit a logistic regression on the true predicates alone ($F1_{LR}$; not an upper bound on their information). On CTU-IoT-23 the model scores 0.93, its head on the true predicates 0.46–0.49 and the logistic regression 0.58; on CIC-IoT-2023, 0.82, 0.62–0.65 and 0.72 (Table S19). Under the evaluated linear classifiers the declared predicates therefore do not account for the CBM's accuracy; the learned concept values carry decision-relevant information that the declared labels do not. Because the intervention changes the head's input distribution, the gap is not a decomposition of accuracy. It shrinks only slightly with γ (CTU 0.476 $\rightarrow$ 0.441, about one seed std).

TABLE S19: Concept-fidelity intervention (JointCBM, five seeds). $F1_{pred}$: normal; $F1_{gt}$: true predicates through the learned classifier; $F1_{LR}$: logistic regression on true predicates alone (deterministic, so no seed std). Mean (std) over five seeds.

Data	γ	$F1_{pred}$	$F1_{gt}$	gap
CTU	0	0.933 (.000)	0.457 (.024)	0.476 (.024)
CTU	0.1	0.933 (.000)	0.489 (.029)	0.444 (.029)
CTU	0.5	0.933 (.000)	0.483 (.016)	0.450 (.015)
CTU	1.0	0.933 (.000)	0.492 (.038)	0.441 (.038)
declared predicates alone (logistic regression): F1 = 0.575				
CIC	0	0.818 (.001)	0.624 (.021)	0.194 (.021)
CIC	0.1	0.819 (.001)	0.642 (.040)	0.177 (.040)
CIC	0.5	0.818 (.001)	0.645 (.028)	0.173 (.028)
CIC	1.0	0.818 (.001)	0.639 (.035)	0.179 (.036)
declared predicates alone (logistic regression): F1 = 0.718				